

Cooling Solutions

It's summer, and like you, your computer, too, needs to stay cool!

I could blame it on the Greenhouse effect, but I am not an environmentalist! Anyone who uses a PC and claims to be an environmentalist is fooling himself. But I am digressing even before getting to the point (and that is a first even for me!).

Problem is, the heat is getting to me—and the evidence is pretty clear. Which brings me to this month's topic—cooling solutions.

I hate summers. The heat and humidity drain me completely, and at times leaves me wondering what life in Rajasthan would be like. It is this time of year that I take the maximum care of my PC too—a ritual I started because I've lost too many PC components during the summers. My experience apart, lately, my mailbox has been bombarded with requests for cooling solutions, and I thought this was the best time to address the heating problems so many face.

In PCs, air is used as the medium for transferring heat from the components to the atmosphere. In fact, until some years ago, liquid cooling for PCs was unheard of, and only recently has liquid cooling emerged as an effective alternative to traditional methods.

To achieve maximum cooling efficiency, you need to manage on the macro (system) as well as micro (component) levels. The perfect synergy between system and component cooling leads to the best overall cooling of your PC.

When it comes to system-level cooling, a lot depends of the PC case (cabinet) you buy. I have often received letters where users say they have two-fan cabinets, but the cooling is not up to the mark. The culprit is often the types of fans and their placement in the case.

Most cabinets in the market today have provision for 80 mm fans—one at the front lower bottom and other at the rear. Instead of opting for such cases, consider a case with provision for one 120 mm fan at the rear. These fans run at a lower speeds, but dispel a high volume of air and so, in general, are more efficient than the faster-spinning 80 mm fans. Moreover, with the launch of the new Prescott-core Pentium IV processors, Intel has recommended changes in the case design, and these cases are called 38 degree cases. There are ducts on the side panel that open straight on the top of the processor and the graphics card for better cooling.

Not many vendors sell them except for those made by Antec and VIP. Cabinets from these brands are expensive and not for the average user.

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However, commonly available brands such as Mercury and iBox have also introduced some good models. Mercury has two series—the KOB and the KM. If you are buying a new PC, I would recommend the new KM series; it conforms to Intel's recommendations. So what if it does not have 120 mm fans?

iBox, too, has the i531 model specifically made in keeping with Intel's recommendations. Zebronics is a popular brand in south India, and has some good models. Such cases, though, will set you back by around Rs 1.5K. Although manufactured conforming to Intel's specs, they work great with AMDs too!

Gamers, though, with high-end processors and graphics cards (nVidia 6800 GT), have no option but to spend that extra cash on an Antec case.

On the component level, the processor, graphics card and PSU dissipate significant amounts of heat as compared to the other components.

Processors are generally provided with a heat sink-fan combo (HSF), and can take care of heating issues under standard working conditions. However, if you overclock the processor, you might need some help keeping the temperature under control.

For processors, three types of solutions are available—the first is the standard HSF, but with better heat conducting materials such as copper in place of aluminium alloy. Such solutions are available from Thermaltake, Cooler Master, AVC and others, and are generally in the range of Rs 1K to Rs 1.5K.

To further improve thermal conductivity, manufacturers introduced a heat pipe—a pipe containing coolant—between the fins of the HSF. Such solutions cost upwards of Rs 2.5K.

The third type are the liquid cooling kits from Thermaltake and Cooler Master. Here, a small pump drives water (or coolant) between a copper block and a radiator. Liquids are more efficient in conducting heat and hence, better cooling is achieved via these kits. Such a kit could set you back by Rs 8.5K. Unfortunately, there are very few shops dealing in such specialised products and hence, the best way to get them would be by importing them.

There are liquid cooling kits available for graphics cards too, but they are extremely rare in the Indian market.

As for PSUs, just buy a good-quality one—it will come with two fans that will keep the heat under control.



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